

Six Functors and Gestalten

Talk 10 — Gestalten Seminar

after [Sch26], Lecture IX (*up to the extension to stacks*)

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Goal of the talk

Starting from a (presentable) six-functor formalism $D: \text{Span}(C) \rightarrow 1\text{Pr}$, we construct *canonically* a functor

$$[-]_D: C \longrightarrow \text{Gest} := \text{StRing}^{\text{op}}, \quad X \longmapsto [X]_D = \text{Gest}(A_{D,X}),$$

the *transmutation* of D , through which D factors as “geometry + quasicoherent sheaves”. The engine is the *category of kernels* $\mathbb{K}_D$; iterating it produces the Stefanich ring $A_{D,X}$. We show $[-]_D$ commutes with finite limits, that every $[f]_D$ is 1-étale and 1-proper, and that $\mathcal{O}(-)_1 \circ [-]_D$ recovers the $(f^*, \otimes)$ -part of D (Proposition 3.3). We stop just before the extension to stacks.

Conventions

Category = ∞ -category. C is a small category with finite limits; $D: \text{Span}(C) \rightarrow 1\text{Pr}$ is a lax symmetric monoidal functor, i.e. a *presentable six-functor formalism*; 1Pr denotes presentable categories and $2\text{Pr} = \text{Mod}_{1\text{Pr}}(1\text{Pr})$; $\mathbb{K}_D = \mathbb{K}_D$ is the *category of kernels* (recalled in Appendix A); $\text{Gest} = \text{StRing}^{\text{op}}$. We write $\mathbb{1}$ for the monoidal unit and $\mathcal{O}(-)_1$ for “global sections / quasicoherent sheaves” (the bottom of a Stefanich ring). Following Appendix A, a morphism is called *D-suave*, *D-prim*, *D-smooth*, *D-étale*, or *D-proper* when it has the corresponding property for the formalism D ; we reserve the unprefix adjectives (smooth, étale, proper, ...) for the underlying geometric notions.

1 From motives to six-functor formalisms

In algebraic geometry there are many cohomology theories: the ℓ -adic ones, one for each prime ℓ ; crystalline cohomology; de Rham and Betti cohomology; and others still. Grothendieck envisioned a *universal* system of coefficients that would dictate all the rest — an initial cohomology theory out of which every other one is read off. In its original, abelian-categorical form, the idea asks for a category Mot_k of *motives*, together with an assignment $X \mapsto \text{Mot}(X)$ (the *motive* of X), such that

every reasonable “cohomology functor” $h^\bullet: \text{Sch}_k^{\text{op}} \rightarrow \mathbb{A}$ valued in an abelian category $\mathbb{A}$ factors through it:

$$\begin{array}{ccc} \text{Sch}_k^{\text{op}} & \xrightarrow{\text{Mot}} & \text{Mot}_k \\ & \searrow h^\bullet & \downarrow \\ & & \mathbb{A}. \end{array}$$

The dashed leg is a *realization*: each Weil cohomology — Betti, de Rham, ℓ -adic — is one realization of the universal motive. All the variation *among* cohomology theories is carried by this choice of realization, while the geometric input $X \mapsto \text{Mot}(X)$ stays fixed. This shape of factorization — a fixed universal object, recovered into each concrete theory by a single leg — is the one we will keep meeting; the goal of the talk is its modern, categorified incarnation.

The first move is to categorify the coefficients. Studying a category C of geometric objects — say schemes of finite type over $\mathbb{Z}$ — on its own is hard; one probes it through linear invariants. The bare association $X \mapsto H^\bullet(X)$ is coarse, since many non-isomorphic spaces share the same cohomology. One remembers far more by attaching to each X a whole category of *coefficients*: *local systems* $\mathcal{L}$, which themselves carry cohomology $H^\bullet(X, \mathcal{L})$, transported along a map $f: Y \rightarrow X$ by fiberwise cohomology $f_*\mathcal{L}$ and — to allow arbitrary maps, not just smooth proper ones — enlarged into a category of *sheaves*. Cohomology then acquires a dual companion, *cohomology with compact support* $H_c^\bullet$, already visible in Poincaré duality.

The structure that packages all of this is a *six-functor formalism* (defined carefully in Section A). Informally, it assigns to each $X \in C$ a symmetric monoidal *category of coefficients* $D(X)$ — in our setting $D(X) \in \mathbf{1Pr}$ — and to each map $f: Y \rightarrow X$ two adjoint pairs

$$f^* \dashv f_*, \quad f_! \dashv f^!,$$

encoding pullback, pushforward (cohomology), exceptional pushforward (cohomology with compact support), and exceptional pullback, all tied to the tensor product $\otimes$ by base change and a projection formula.

We now look for the categorified analogue of Mot_k : a fixed target through which *every* six-functor formalism factors. A first candidate is the category AnStck of *analytic stacks*, through which the $(f^*, \otimes)$ -part $D^{*, \otimes}$ often factors. Grothendieck and Scholze are then after the same thing — a universal coefficient through which every theory factors — and differ only in *which leg of the triangle they hold fixed*:

	Grothendieck: motives	Scholze: six functors
factorization	$\begin{array}{ccc} \mathrm{Sch}_k^{\mathrm{op}} & \xrightarrow{\mathrm{Mot}} & \mathrm{Mot}_k \\ & \searrow h^* & \downarrow \mathrm{real.} \\ & & \mathbb{A} \end{array}$	$\begin{array}{ccc} C^{\mathrm{op}} & \xrightarrow{X \mapsto X^?} & \mathrm{AnStck}^{\mathrm{op}} \\ & \searrow D^{*,\otimes} & \downarrow D_{qc} \\ & & \mathrm{CAlg}(\mathrm{1Pr}) \end{array}$
universal coefficient	Mot_k	AnStck (later Gest)
front leg	$\mathrm{Mot} \text{ --- fixed}$	transmutation $X \mapsto X^? \text{ --- varies with } D$
recovering leg	realization $\mathrm{Mot}_k \rightarrow \mathbb{A} \text{ --- varies}$	$D_{qc} = O(-)_1 \text{ --- fixed}$
variation lives in	the realization	the transmutation

The same triangle, with the fixed and varying legs interchanged. Scholze packages all the realizations into the single theory-independent leg $O(-)_1 = D_{qc}$ — the same for every D — and studies the front leg instead: the *transmutation* $X \mapsto X^?$, after Bhatt [Bha22, Remark 2.3.8], which converts a geometric object into its universal coefficient and commutes with finite limits (equivalently, its opposite commutes with finite colimits). So the classical Betti and de Rham realizations are just $O(-)_1$ applied to the transmutations by the Betti and de Rham ring stacks; the dashed top leg is the one caveat — AnStck achieves this only for *some* D .

For schemes (separated of finite type over a field, say) such transmutations are produced from a *ring stack*, by the following criterion of Scholze.

Theorem 1.1 ([Sch25, Theorem 10.6]). *Work in the category of analytic stacks over some base analytic stack S . Let k be a ring and let R be a k -algebra stack satisfying the following conditions.*

- (1) *The map $f: R \rightarrow *$ is $!$ -able and D -smooth, and the map $1 \rightarrow f_! f^! 1$ is an equivalence.*
- (2) *The origin $i: * \xrightarrow{0} R$ is a closed immersion, the units $j: R^\times \hookrightarrow R$ form its open complement, and*

$$j_! 1 \rightarrow 1 \rightarrow i_* 1$$

is a fiber sequence in $D_{qc}(R)$.

This induces a transmutation functor

$$X \mapsto X_R$$

taking any separated scheme X of finite type over k to the static (i.e. 0-truncated) analytic stack over S given by the sheafification of the functor

$$\mathrm{AnRing}_S \ni A \mapsto X(R(A)).$$

The functor $X \mapsto X_R$ commutes with finite limits and Zariski gluing. Moreover, the functor $X \mapsto X_R$ sends étale maps to D -étale maps, proper maps to D -proper maps, and smooth maps to D -smooth maps, and thus $X \mapsto D_{qc}(X_R)$ yields a six-functor formalism with these properties.

For instance, the Betti stack and the solid de Rham stack arise from such ring stacks.

The fixed second leg D_{qc} satisfies the *categorical Künneth formula*: by [Kes25], on analytic stacks over an affine base the external product is an equivalence.¹ Hence, if D factors as above, then D itself satisfies categorical Künneth:

$$D(X) \otimes_{D(*)} D(Y) \xrightarrow{\sim} D(X \times Y) \quad \text{for all } X, Y.$$

For SH, for IndCoh, and for ℓ -adic sheaves, however, the categorical Künneth formula *fails* — so none of these factors through AnStck in this way.

Remark 1.2 (Why the Künneth formula fails for SH, IndCoh, and ℓ -adic sheaves). In each case the external product $D(X) \otimes_{D(*)} D(Y) \xrightarrow{\boxtimes} D(X \times Y)$ fails to be an equivalence — it is not essentially surjective — so the kernel category $\mathbb{K}_D$ is strictly larger than $\text{Mod}_{D(*)}(1\text{Pr})$ (Theorem 2.3). (For ℓ -adic sheaves the external product is moreover fully faithful [Gai25]; for IndCoh and SH the cited sources assert only non-equivalence.)

- *ℓ -adic sheaves* [Gai25]. The functor $\text{Shv}(Y) \otimes \text{Shv}(Y') \xrightarrow{\boxtimes} \text{Shv}(Y \times Y')$ is *never* an equivalence once Y, Y' are positive-dimensional of finite type. For a witness take $Y' = Y$ and the diagonal $\Delta: Y \hookrightarrow Y \times Y$: the constructible sheaf $\Delta_* \mathbb{1}_Y$ is compact, with singular support the conormal $N_\Delta^* \subset T^*(Y \times Y)$. Every external product satisfies $\text{SS}(F \boxtimes G) \subseteq \text{SS}(F) \times \text{SS}(G)$ [KS90, §5.4], while N_Δ^* is not contained in any finite union of products of conic Lagrangians (the diagonal $Y \subset Y \times Y$ is not a product). Hence $\Delta_* \mathbb{1}_Y$ is not a retract of a finite colimit of external products, so it escapes the image. (For ℓ -adic coefficients, singular support is Beilinson’s [Bei16].)
- *Ind-coherent sheaves* [Ste25]. For IndCoh on schemes almost of finite presentation over k , the categorification tower reaches module form only late: for an arbitrary map $X \rightarrow Y$ one has $n\text{-IndCoh}(X) \simeq \text{Mod}_{(n-1)\text{-IndCoh}(X/Y)}(n\text{-IndCoh}(Y))$ for $n \geq 3$ (the module form already appears at level 2 for closed immersions), and if k is perfect then $\text{Pr}_{\text{IndCoh}} \simeq \text{Pr}$. It is the bottom level that fails, and this is precisely the failure of Künneth: over a base S the formalism $X \mapsto \text{IndCoh}(X)$ takes values in $1\text{Pr}_S = \text{Mod}_{\text{QCoh}(S)}(1\text{Pr})$, the 2-category of $\text{QCoh}(S)$ -linear presentable categories (= 2-QCoh(S)), and its right-lax monoidal structure is the external product

$$\text{IndCoh}(X_1) \otimes_{\text{QCoh}(S)} \text{IndCoh}(X_2) \longrightarrow \text{IndCoh}(X_1 \times_S X_2).$$

As Gaitsgory–Rozenblyum–Varshavsky note [GRV25], even when S, X_1, X_2 are smooth this is *not* an equivalence as soon as $X_1 \times_S X_2$ fails to be smooth — IndCoh agreeing with QCoh only on smooth targets, so that Stefanich’s 2-IndCoh(S) properly enlarges 1Pr_S . (Contrast QCoh, which *does* satisfy Künneth on perfect stacks [BFN10, Thm. 4.14].)

¹Precisely, [Kes25, Cor. 1.5.1]: over a base with representable diagonal admitting an affine $!$ -cover — in particular over an affine base — every $!$ -able object is Künneth. Over a general stacky base the *relative* formula may fail, but that is not what is at stake here.

- *Motivic spectra* [Aok26]. Aoki shows the kernel 2-category $\mathbb{K}_{\text{SH}}$ of SH is *freely* generated by a ring stack that is homologically trivial, smooth, and sutured — and is emphatically not the unit, giving a moduli description of the genuinely 2-categorical gestalt of SH. Were Künneth to hold, $\mathbb{K}_{\text{SH}}$ would collapse to the single floor $\text{Mod}_{\text{SH}(\ast)}(1\text{Pr})$.

This is the gap we set out to fill. We enlarge the rigid target AnStck to the category of *Gestalten* $\text{Gest} = \text{StRing}^{\text{op}}$, and factor *every* presentable six-functor formalism D through a canonically associated transmutation:

$$\begin{array}{ccc} C^{\text{op}} & \xrightarrow{[-]_D} & \text{Gest}^{\text{op}} \\ & \searrow \scriptstyle D^*, \otimes & \downarrow \scriptstyle O(-)_1 \\ & & \text{CAlg}(1\text{Pr}). \end{array}$$

In contrast with the analytic-stack picture, which applies only to *some* D , this construction — following Scholze [Sch26] and Stefanich [Ste25] — works for *every* presentable D . *Gestalten* is in this sense the *universal coefficient category* for presentable six-functor formalisms, the categorified counterpart of Grothendieck’s Mot_k : quasicoherent sheaves on Gest — the fixed functor $O(-)_1$ — is the single, theory-independent recovering leg, the *packaged realization*, through which every 1Pr -valued D factors. The price is that $[X]_D$ is now a genuinely higher-categorical object, obtained by iterating the category of kernels $\mathbb{K}_D$ into a *Stefanich ring* $A_{D,X}$. The very failure of Künneth that obstructed the factorization through AnStck is exactly what the higher target absorbs (Theorem 2.3).

The whole picture is one *ladder of categorical levels*: each invariant of X records data one categorical degree higher than the last — and so supports correspondingly richer algebraic operations — while the transmutation $[-]_D$ records every level at once.

Level	Invariant $X \mapsto$	n -categorical algebra	Available operations
0	$H^\bullet(X)$	cohomology ring (numbers)	$+, \times$
1	$D(X)$	symmetric monoidal category	limits, colimits, $\otimes$
2	$\mathbb{K}_{D,X}$	category of kernels	limits, colimits, $\otimes$, adjoints, monads
∞	$[X]_D$	Stefanich ring (a Gestalt)	all of the above, at every level

2 The category of kernels $\mathbb{K}_D$

We now construct the finite-limit-preserving transmutation $[-]_D: C \rightarrow \text{Gest}$ at the heart of the universal-coefficient picture above. The most naive candidate uses D directly — send each X to the Gestalt of its coefficient category, i.e. compose

$$C \hookrightarrow \text{Span}(C) \xrightarrow{D} 1\text{Pr} \longrightarrow \text{Gest}, \quad X \mapsto \text{Gest}(D(X))$$

(precisely, the $(f^*, \otimes)$ -part $D^{*, \otimes}: C^{\text{op}} \rightarrow \text{CAlg}(\mathbf{1Pr})$ postcomposed with the bottom-floor inclusion $\text{CAlg}(\mathbf{1Pr}) \hookrightarrow \text{StRing} = \text{Gest}^{\text{op}}$). But this need not preserve pullbacks: the product $X \times Y$ is sent to $\text{Gest}(D(X \times Y))$, whereas the product of $\text{Gest}(D(X))$ and $\text{Gest}(D(Y))$ in Gest is $\text{Gest}(D(X) \otimes_{D(*)} D(Y))$ — products in $\text{Gest} = \text{StRing}^{\text{op}}$ are relative tensor products of coefficients. The two coincide exactly when D satisfies the *categorical Künneth formula* $D(X) \otimes_{D(*)} D(Y) \xrightarrow{\sim} D(X \times Y)$, which in general fails (Theorem 2.3). The *category of kernels* $\mathbb{K}_D$ is the universal fix — it enlarges the coefficients into a theory for which Künneth holds one categorification up, so that iterating it into the Stefanich ring (Section 3) yields a transmutation that genuinely preserves finite limits.

The kernel 2-category is constructed in Appendix A (Definition A.9); here $\mathbb{K}_D = \mathbb{K}_{D,*}$ is its absolute case, over the terminal object of C .

Definition 2.1. Let C be a category with finite limits. Regarding the six-functor formalism D as a commutative algebra object in $\text{Fun}(\text{Span}(C), \mathbf{1Pr})$, equipped with the Day convolution symmetric monoidal structure, the 2-category of kernels of D is the module category

$$\mathbb{K}_D := \text{Mod}_D(\text{Fun}(\text{Span}(C), \mathbf{1Pr})).$$

This comes with a symmetric monoidal functor from $\text{Span}(C)^{\text{op}}$, given as the composite

$$\text{Span}(C)^{\text{op}} \rightarrow \text{Fun}(\text{Span}(C), \mathbf{1Pr}) \rightarrow \text{Mod}_D(\text{Fun}(\text{Span}(C), \mathbf{1Pr})) = \mathbb{K}_D,$$

namely the Yoneda embedding followed by the free D -module functor $D \otimes (-)$. We denote it by $X \mapsto [X]_{\mathbb{K}_D}$. Since $[X]_{\mathbb{K}_D}$ is the free D -module on the representable presheaf at X , it is easy to map out of: by the free–forgetful adjunction composed with the Yoneda lemma, for any $X \in \text{Span}(C)^{\text{op}}$ and $M \in \mathbb{K}_D$ one has

$$\text{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, M) = M(X).$$

In particular, as the unit object of $\mathbb{K}_D$ is D itself, taking $M = \mathbb{1}$ gives

$$\text{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, \mathbb{1}) = D(X).$$

Now every object of $\text{Span}(C)$ is canonically self-dual ([HM24, p. 2.3.9]), and the embedding is symmetric monoidal, so $[Y]_{\mathbb{K}_D} \simeq [Y]_{\mathbb{K}_D}^{\vee}$ and $[X]_{\mathbb{K}_D} \otimes [Y]_{\mathbb{K}_D} \simeq [X \times Y]_{\mathbb{K}_D}$. Hence in general

$$\text{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, [Y]_{\mathbb{K}_D}) \simeq \text{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D} \otimes [Y]_{\mathbb{K}_D}^{\vee}, \mathbb{1}) \simeq \text{Hom}_{\mathbb{K}_D}([X \times Y]_{\mathbb{K}_D}, \mathbb{1}) = D(X \times Y),$$

which recovers Definition A.9 (there the mapping object is written $\text{Fun}_Y(X_1, X_2) = D(X_1 \times_Y X_2)$; here Y is the terminal object, so $\text{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, [Y]_{\mathbb{K}_D})$ is the absolute case $\text{Fun}_*(X, Y)$). In fact, the two constructions can be compared by the following theorem:

Theorem 2.2 (Aoki, [Aok26, Theorem 2.6]). *Let S be a small closed symmetric monoidal category and $V \in \text{CAlg}(\mathbf{Pr}^{\text{L}})$. Let $D: S \rightarrow V$ be a lax symmetric monoidal functor. Then there is a canonical equivalence*

$$\text{PSh}^V(\tilde{S}^{\text{op}}) \simeq \text{Mod}_D(\text{Fun}(S, V)),$$

where $\tilde{S}$ is the V -enriched category obtained by base changing the self-enrichment of S along D .

Proof. Aoki credits this argument to H. Heine; we expand it, isolating the monadicity step, which is the monadicity theorem for presentable module categories of Reutter–Zetto [RZ25, Theorems 4.7 and 4.12].

Setup. Since $V \in \mathbf{CAlg}(\mathbf{Pr}^{\mathbf{L}})$, write $U: \mathbf{An} \rightarrow V$ for the unit functor, i.e. the symmetric monoidal left adjoint sending the point to the monoidal unit $\mathbb{1}_V$. We first observe that $\mathbf{Fun}(S, V)$ with its pointwise structure is exactly the category of V -valued presheaves on S for the *trivial* V -enrichment S^{triv} of S : that is, $\mathbf{Fun}(S, V) = \mathbf{PSh}^V(S^{\text{op}})$, where S^{triv} has the same objects as S and mapping objects $U(\mathbf{Map}_S(\mathbb{1}, -))$ obtained by pushing the mapping anima $\mathbf{Map}_S(\mathbb{1}, -): S \rightarrow \mathbf{An}$ into V along U .

Change of enrichment. Because S is closed symmetric monoidal, $U(\mathbf{Map}_S(\mathbb{1}, -))$ is the unit of $\mathbf{CAlg}(\mathbf{Fun}(S, V))$, so there is a canonical morphism

$$U(\mathbf{Map}_S(\mathbb{1}, -)) \longrightarrow D$$

in $\mathbf{CAlg}(\mathbf{Fun}(S, V))$. Base changing the self-enrichment of S first along $U(\mathbf{Map}_S(\mathbb{1}, -))$ and then along D , this morphism induces a V -enriched, identity-on-objects functor

$$S^{\text{triv}} \longrightarrow \tilde{S},$$

the source being the trivially enriched copy of S and $\tilde{S}$ the D -base-changed enrichment of the theorem. Write f for its opposite. Passing to enriched presheaves yields an adjunction

$$f_! \dashv f^* \quad \text{in } \mathbf{Mod}_V(\mathbf{Pr}^{\mathbf{L}}),$$

in which $f^*: \mathbf{PSh}^V(\tilde{S}^{\text{op}}) \rightarrow \mathbf{PSh}^V(S^{\text{op}}) = \mathbf{Fun}(S, V)$ is the restriction of enrichment and $f_!$ is its V -linear left adjoint (left Kan extension).

Monadicity. The restriction f^* is the right adjoint of the Eilenberg–Moore (free/forgetful) adjunction attached to the change-of-enrichment functor: its left adjoint sits in the image of the fully faithful Eilenberg–Moore embedding, and f^* is V -linear, colimit-preserving and conservative. By the monadicity theorem for presentable module categories [RZ25, Theorems 4.7 and 4.12], f^* is therefore monadic in $\mathbf{Mod}_V(\mathbf{Pr}^{\mathbf{L}})$. Writing $T := f^* f_!$ for the resulting V -linear monad on $\mathbf{Fun}(S, V)$, monadicity gives an equivalence

$$\mathbf{PSh}^V(\tilde{S}^{\text{op}}) \simeq \mathbf{Mod}_T(\mathbf{Fun}(S, V)).$$

It thus remains to identify the monad T .

Identification of the monad. The monad $T = f^* f_!$ is V -linear and preserves colimits, hence as an endofunctor of $\mathbf{Fun}(S, V)$ it is determined by its value on the unit: $T \simeq T(\mathbb{1}) \otimes (-)$. Now $f_!$ is symmetric monoidal, so $f_! \mathbb{1} \simeq \mathbb{1}$, while by construction the restriction satisfies $f^* \mathbb{1} \simeq D$; hence

$$T(\mathbb{1}) = f^* f_! \mathbb{1} \simeq f^* \mathbb{1} \simeq D,$$

and the unit map $\mathbb{1} \rightarrow T$ identifies D with the unit of T , yielding a morphism of monads $D \rightarrow f^* f_!$. Consequently $T \simeq - \otimes D$ as monads, and its algebras are precisely the D -modules. By linearity it suffices to check this equivalence on representables, which is immediate. Combining with the displayed equivalence,

$$\mathrm{PSh}^V(\tilde{S}^{\mathrm{op}}) \simeq \mathrm{Mod}_T(\mathrm{Fun}(S, V)) \simeq \mathrm{Mod}_D(\mathrm{Fun}(S, V)),$$

as claimed. $\square$

Now, take $V = \mathbf{1Pr}$ and $S = \mathrm{Span}(C)$, we obtain the desired result.

It remains to show that $\mathbb{K}_D$ satisfies categorical Künneth formula (i.e. $C \rightarrow \mathrm{Gest}$ preserves pull-backs).

Proposition 2.3 ([Sch26, Prop. 9.3]). *The natural functor*

$$\mathrm{Mod}_{D(*)}(\mathbf{1Pr}) \longrightarrow \mathbb{K}_D$$

is fully faithful, with essential image the “restricted²” objects: $M \in \mathbb{K}_D$ lies in it iff $D(X) \otimes_{D()} M(*) \rightarrow M(X)$ is an equivalence for all X . In particular,*

$$\mathbb{K}_D = \mathrm{Mod}_{D(*)}(\mathbf{1Pr}) \iff D(X) \otimes_{D(*)} D(Y) \xrightarrow{\sim} D(X \times Y) \text{ for all } X, Y.$$

Proof. We work with the description $\mathbb{K}_D = \mathrm{Mod}_D(\mathrm{Fun}(\mathrm{Span}(C), \mathbf{1Pr}))$ of Theorem 2.1 and the formula $\mathrm{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, M) = M(X)$ established there.

Evaluation is atomic. For each X , that formula says the corepresentable of $[X]_{\mathbb{K}_D}$ is the evaluation functor

$$\mathrm{ev}_X = \mathrm{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, -) : \mathbb{K}_D \longrightarrow \mathbf{1Pr}, \quad M \longmapsto M(X).$$

Colimits in $\mathbb{K}_D = \mathrm{Mod}_D(\mathrm{Fun}(\mathrm{Span}(C), \mathbf{1Pr}))$ are created by the forgetful functor to $\mathrm{Fun}(\mathrm{Span}(C), \mathbf{1Pr})$, where they are computed pointwise; hence ev_X preserves all colimits, and it is plainly $\mathbf{1Pr}$ -linear. Thus every $[X]_{\mathbb{K}_D}$ is $\mathbf{1Pr}$ -atomic. In particular the unit $\mathbb{1} = [*]_{\mathbb{K}_D}$ is $\mathbf{1Pr}$ -atomic, with $\mathrm{End}_{\mathbb{K}_D}(\mathbb{1}) = \mathrm{ev}_*(\mathbb{1}) = D(*)$.

The natural functor and its right adjoint. Since $\mathrm{ev}_*(M) = M(*)$ is naturally a module over $\mathrm{End}_{\mathbb{K}_D}(\mathbb{1}) = D(*)$, the functor ev_* refines to a colimit-preserving functor

$$G : \mathbb{K}_D \longrightarrow \mathrm{Mod}_{D(*)}(\mathbf{1Pr}), \quad M \longmapsto M(*).$$

The functor of the statement is its left adjoint — induction along $D(*) = \mathrm{End}_{\mathbb{K}_D}(\mathbb{1}) \rightarrow \mathbb{K}_D$,

$$L : \mathrm{Mod}_{D(*)}(\mathbf{1Pr}) \longrightarrow \mathbb{K}_D, \quad N \longmapsto N \otimes_{D(*)} \mathbb{1},$$

²in the sense of [Gai25]

the unique 1Pr-linear colimit-preserving functor with $L(D(*)) = \mathbb{1}$; one has $L \dashv G$.

Full faithfulness. L is fully faithful iff the unit $N \rightarrow GL(N)$ of the adjunction is an equivalence for every N . Here

$$GL(N) = (N \otimes_{D(*)} \mathbb{1})(*) \simeq N \otimes_{D(*)} \mathbb{1}(*) = N \otimes_{D(*)} D(*) = N.$$

The middle equivalence is the crux, and it is exactly where atomicity enters: ev_* is colimit-preserving and 1Pr-linear, so it commutes with the relative tensor product $N \otimes_{D(*)} (-)$ (a geometric realization of the bar construction); and $\mathbb{1}(*) = D(*)$. Hence $GL \simeq \text{id}$ and L is fully faithful.

The essential image. As L is fully faithful with right adjoint G , an object $M \in \mathbb{K}_D$ lies in the essential image of L iff the counit $\varepsilon_M: LG(M) \rightarrow M$ is an equivalence. Evaluating $LG(M) = M(*) \otimes_{D(*)} \mathbb{1}$ at X — again using that ev_X commutes with $M(*) \otimes_{D(*)} (-)$, together with $\mathbb{1}(X) = D(X)$ —

$$(LG(M))(X) \simeq M(*) \otimes_{D(*)} \mathbb{1}(X) = D(X) \otimes_{D(*)} M(*),$$

and ε_M at X is precisely the canonical map $D(X) \otimes_{D(*)} M(*) \rightarrow M(X)$. So M is in the image — i.e. *restricted* — iff this map is an equivalence for all X , as claimed.

The Künneth criterion. The restricted objects are closed under colimits: LG and id both preserve colimits, so the locus where the natural transformation ε is invertible is closed under colimits. Since $\mathbb{K}_D$ is generated under colimits by the representables $[Y]_{\mathbb{K}_D}$, the functor L is essentially surjective — equivalently an equivalence, i.e. $\mathbb{K}_D = \text{Mod}_{D(*)}(1\text{Pr})$ — iff every $[Y]_{\mathbb{K}_D}$ is restricted. Finally, by the $\text{Hom}_{\mathbb{K}_D}$ -formula,

$$[Y]_{\mathbb{K}_D}(*) = \text{Hom}_{\mathbb{K}_D}(\mathbb{1}, [Y]_{\mathbb{K}_D}) = D(Y), \quad [Y]_{\mathbb{K}_D}(X) = \text{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, [Y]_{\mathbb{K}_D}) = D(X \times Y),$$

so $[Y]_{\mathbb{K}_D}$ is restricted iff $D(X) \otimes_{D(*)} D(Y) \rightarrow D(X \times Y)$ is an equivalence for all X . Ranging over Y as well, this is exactly the categorical Künneth formula. $\square$

3 Transmutation

Now, we can try to construct the transmutation functor $C^{\text{op}} \rightarrow \text{Gest}^{\text{op}}$

Proposition 3.1 ([Sch26, Prop. 9.4]). *For every $f: Y \rightarrow X$ the functor $f_1^*: \mathbb{K}_{D,X} \rightarrow \mathbb{K}_{D,Y}$ admits isomorphic left and right adjoints $f_{1!} \cong f_{1*}$, whose formation commutes with base change.*

[TODO: Proof sketch (the forgetful $C_{/Y} \rightarrow C_{/X}$ gives an ambidextrous adjoint to $\text{Span}(C_{/X}) \rightarrow \text{Span}(C_{/Y})$, passing through modules over D). Then: $X \mapsto \text{Gest}(\mathbb{K}_{D,X})$ is 2-affine, so 2-proper and 1-prim; 1-suaveness is *not* clear (left adjoints unknown), and upgrading to 1-étale/1-proper via diagonals needs the Künneth formula, which may fail — this forces the iteration.]

[TODO: The iteration: $X \mapsto \mathbb{K}_{D,X}$ admits base-change-compatible left adjoints, upgrading to a 6FF $\text{Span}(C) \rightarrow 2\text{Pr}$; iterating gives the *Stefanich ring* (Stefanich, [Ste25]).]

$$A_{D,X} = (D(X), \mathbb{K}_{D,X}, \dots) \in \text{StRing}, \quad \text{functorial in } C^{\text{op}} \rightarrow \text{StRing}.$$

Definition 3.2 (Transmutation). With $\text{Gest} := \text{StRing}^{\text{op}}$, the *transmutation* of $\mathbf{D}$ is

$$[-]_{\mathbf{D}} : C \longrightarrow \text{Gest}/_{\text{Gest}(A_{\mathbf{D}})}, \quad X \longmapsto [X]_{\mathbf{D}} = \text{Gest}(A_{\mathbf{D},X}).$$

Theorem 3.3 ([Sch26, Prop. 9.5]). *For every presentable six-functor formalism $\mathbf{D}$, the transmutation $[-]_{\mathbf{D}}$ satisfies:*

- (1) *it commutes with finite limits;*
- (2) *for each $f : Y \rightarrow X$ in C , the map $[f]_{\mathbf{D}} : [Y]_{\mathbf{D}} \rightarrow [X]_{\mathbf{D}}$ is 1-étale and 1-proper, with $f_{1!} \cong f_{1*}$ (the left and right adjoints to the kernel pullback $f_1^* : \mathbb{K}_{\mathbf{D},X} \rightarrow \mathbb{K}_{\mathbf{D},Y}$ agreeing);*
- (3) *the composite $O(-)_1 \circ [-]_{\mathbf{D}}$ recovers the $(f^*, \otimes)$ -functoriality of $\mathbf{D}$.*

Moreover the construction is, in Scholze’s words, “completely canonical”.

[TODO: Proof sketch: each level is 1-prim $\wedge$ 1-suave by Theorem 3.1 (n -suave extra condition automatic by ambidexterity); finite limits reduce to binary products = Künneth for lower-shriek, using base-change compatibility of the right adjoints.]

Warning 3.4. $O(-)_1 \circ [-]_{\mathbf{D}}$ recovers $\mathbf{D}$ only at the level of $(f^*, \otimes)$; the shriek functors are not seen by the target $\text{CAlg}(1\text{Pr})$. [TODO: Reconstructing the full formalism (the “!-able” maps of Gestalten) is the subject of the appendix to Lecture IX — “not definitive”.]

4 Dictionary: six-functor notions vs. Gestalten

[TODO: Derive the dictionary (the unit $\mathbb{1}_X$ defines the counit $(B/A)_1^! \rightarrow \mathbb{1}$ in $A_2 = \mathbb{K}_{\mathbf{D},Y}$, [Sch26, Def. 6.16]): f suave/prim $\Leftrightarrow [f]_{\mathbf{D}}$ 0-suave/0-prim; f cohomologically étale/proper $\Leftrightarrow f$ truncated $\wedge [f]_{\mathbf{D}}$ 0-étale/0-proper. After transmutation every morphism is automatically 1-étale/1-proper (ambidexterity at layer 1, $f_{1!} \cong f_{1*}$, Theorem 3.1), so only the layer-0 data is non-trivial.]

Remark 4.1 ([Sch26, Rem. 9.6]). In tabular form:

f in C	$[f]_{\mathbf{D}}$ in Gest
suave	0-suave
prim	0-prim
cohomologically étale	truncated $\wedge$ 0-étale
cohomologically proper	truncated $\wedge$ 0-proper

Remark 4.2 (Stack-theoretic perspective). The suave and prim notions above have a clean stack-level form, due to Aoki [Aok26, Def. 4.5]: a morphism $[X] \rightarrow [Y]$ in the kernel 2-category $\mathbb{K}_{\mathbf{D},Y}$

(Theorem A.9) is *weakly suave* if it admits an $[X]$ -linear left adjoint, *prim* if it admits an $[X]$ -linear right adjoint, and *suave* if it is weakly suave and $[Y]$ is dualizable over $[X]$. Aoki’s *weakly suave* matches the object-level Theorem A.11 up to the standard reformulation “is a left adjoint” versus “admits an $[X]$ -linear left adjoint” (via the self-duality $\mathbb{K}_{D,Y} \simeq \mathbb{K}_{D,Y}^{\text{op}}$), while his *suave* adds Aoki’s dualizability of $[Y]$ over $[X]$ — equivalently, by Theorem A.20, of the suave dual — exactly the extra input promoting suaveness to cohomological smoothness. This is the stack-level reading behind the dictionary above: suaveness and primness of f become properties of $[f]_D$ in Gest.

Outlook: towards the extension to stacks

Recall from [Sch25, Def. 5.3] the D -topology on C : a family $\{f_i: Y_i \rightarrow X\}$ (with each f_i in E and forming a cover for the canonical topology) is a D -cover iff it moreover satisfies *universal !-descent*, i.e. after any base change $X' \rightarrow X$ the upper-shriek functor

$$D(X') \xrightarrow{\sim} \lim \left(\prod_i D(Y_i \times_X X') \rightrightarrows \prod_{i,j} D(Y_i \times_X Y_j \times_X X') \cdots \right)$$

is an equivalence.

Proposition ([Sch26, Prop. 9.7]). If $\{f_i: Y_i \rightarrow X\}$ satisfies universal !-descent, then $\coprod_i [Y_i]_D \rightarrow [X]_D$ is a cover (effective epimorphism) in Gest.

Corollary ([Sch26, Cor. 9.8]). Hence $[-]_D$ extends to a functor

$$\text{Shv}_D(C) \longrightarrow \text{Gest}/[-]_D, \quad X \longmapsto [X]_D,$$

from D -sheaves on C (with values in 1-étale Gestalten), commuting with countable colimits and finite limits.

[TODO: Punchline: the D -topology is the restriction along $[-]_D$ of the canonical topology on Gestalten; hand over to Talk 11.]

One-line summary

$D \rightsquigarrow \mathbb{K}_D \rightsquigarrow A_{D,X} \rightsquigarrow [X]_D = \text{Gest}(A_{D,X})$: the six-functor formalism is quasicohherent sheaves on its transmutation, canonically and for all D — so Gest is the universal coefficient category for presentable six-functor formalisms.

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screen use; for this reason I relied throughout on the AI assistant Claude Code to help complete the text.

A Six-functor formalisms, smoothness, and suaveness

A modern viewpoint, originating in Grothendieck’s school, organizes “cohomology with coefficients” on a class of geometric objects C as a single coherent package: an assignment $X \mapsto D(X)$ of a category of “coefficient systems” to each object, together with six functors relating these categories under morphisms.

An informal definition. A **six-functor formalism** on C assigns to each $X \in C$ a presentable symmetric monoidal stable category $D(X)$, and to each morphism $f: Y \rightarrow X$ in C four functors

$$f^* \dashv f_*: D(Y) \rightleftarrows D(X), \quad f_! \dashv f^!: D(X) \rightleftarrows D(Y),$$

together with the symmetric monoidal product $\otimes$ and internal hom $\underline{\text{Hom}}$ on each $D(X)$, such that:

- f^* is colimit-preserving and symmetric monoidal, and $f_!$ is colimit-preserving and satisfies the projection formula;
- base change: for every pullback square in C ,

$$\begin{array}{ccc} Y' & \xrightarrow{g'} & Y \\ f' \downarrow & \lrcorner & \downarrow f \\ X' & \xrightarrow{g} & X \end{array}$$

the canonical maps $g^* f_! \rightarrow f'_! g'^*$ and $f^! g_* \rightarrow g'_* f'^!$ are equivalences;

- projection formula: $f_!(A \otimes f^* B) \simeq f_!(A) \otimes B$ for $A \in D(Y)$, $B \in D(X)$.

Coherent versions of these data live as a 2-functor out of a span 2-category; the span-category formulation is the precise way of encoding all the higher coherences and is the subject of Section A.1.

The six functors carry distinct geometric content:

- f^* is *pullback*, f_* the *direct image*, $\otimes$ the *symmetric monoidal product* encoding cup products, and $\underline{\text{Hom}}$ the right adjoint to $\otimes$.
- $f_!$ is the *exceptional / proper-supported pushforward*: it agrees with f_* when f is proper and with $j_\#$ (extension by zero) when f is an open immersion.

- $f^!$ is the *exceptional pullback*, encoding orientation and duality data; in many examples, for f smooth of relative dimension d , $f^! \simeq f^* \otimes \omega_f$ with ω_f the relative dualizing complex.

The three operations f_* , $f^!$, $\underline{\mathrm{Hom}}$ are *right adjoints* of f^* , $f_!$, $\otimes$ respectively, hence determined by the others. Modern formulations therefore typically specify only the three left functors $(f^*, f_!, \otimes)$ (a so-called **three-functor formalism**) and recover the rest by adjunction.

Three representative examples are:

- *Locally compact Hausdorff topological spaces* (Heyer–Mann [HM24]; see also [Sch25, Lecture VII]): here X ranges over (condensed) topological spaces and the coefficient category $\mathrm{D}(X)$ is the derived category of sheaves of Λ -modules on X , for a fixed coefficient ring Λ .
- *ℓ -adic étale cohomology of schemes* [Sch25, Appendix to Lecture VII].
- *The motivic formalism SH* [Sch25, Lecture XI] — the universal $\mathbb{A}^1$ -invariant one, by Drew–Gallauer.

For pedagogical introductions complementing the present appendix, we recommend Scholze’s lecture notes [Sch25], Gallauer’s mini-course [Gal21] (the source of much of this section’s exposition), and the textbook treatment in Cisinski–Déglise [CD19]. The encoding via span 2-categories adopted here is closest to Mann–Scholze [Man22; Sch25] and Cnossen–Lenz–Linskens [CLL25], to which we turn in the next subsection.

A.1 Span categories and suitable decompositions

The economical encoding of a six-functor formalism, initiated by Liu–Zheng [LZ24] and brought to its present form by Mann [Man22] and Scholze [Sch25], packages all the operations $(f^*, f_!, \otimes)$ and their coherences into a single lax symmetric monoidal functor on a span category. Given a wide subcategory $E \subset C$ closed under composition and base change, the **span category** $\mathrm{Span}(C, E)$ has the same objects as C and morphisms $X \rightarrow Y$ given by spans

$$X \leftarrow Z \xrightarrow{e} Y, \quad e \in E,$$

composed by pullback; when C has finite products, $\mathrm{Span}(C, E)$ inherits a symmetric monoidal structure induced by the finite products of C , with unit the terminal object. (This monoidal structure is *not* Cartesian: the tensor $X \otimes Y \simeq X \times Y$ is induced by the *product* of C , and is in general neither the categorical product nor the coproduct of $\mathrm{Span}(C, E)$. Independently, $\mathrm{Span}(C, E)$ is semiadditive, but its biproduct $X \oplus Y \simeq X \sqcup Y$ comes from the *coproduct* of C — a different operation from the tensor.) A **three-functor formalism** on (C, E) is then a lax symmetric monoidal functor

$$\mathrm{D}: \mathrm{Span}(C, E) \longrightarrow \mathrm{Cat},$$

encoding f^* (backward legs), $f_!$ (forward legs), and $\otimes$ (monoidal structure) in a single coherent package; the right adjoints f_* , $f^!$, $\underline{\text{Hom}}$ are recovered automatically and are therefore *properties* of D rather than additional data.

The challenge is that constructing a lax monoidal functor out of $\text{Span}(C, E)$ directly is difficult. In practice, one starts from a much simpler piece of data, namely a functor

$$D_0: C^{\text{op}} \longrightarrow \text{CAlg}(\text{Cat})$$

encoding only f^* and $\otimes$, and asks when it extends to $\text{Span}(C, E)$. The key input is a **suitable decomposition** of E in the sense of Mann [Man22, Definition A.5.9]: a pair of wide subcategories $I, P \subset E$ (thought of as “étale-like” and “proper-like” morphisms respectively) such that every $e \in E$ factors as $e = p \circ i$ with $i \in I$ and $p \in P$, subject to mild cancellation and truncation axioms. The extension theorem then reads: if $i^* = D_0(i)$ admits a left adjoint $i_{\sharp}$ and $p^* = D_0(p)$ admits a right adjoint p_* satisfying the expected base change and projection formulae—in particular the mixed Beck–Chevalley equivalence $i_{\sharp}p'_* \simeq p_*i'_{\sharp}$ —then D_0 admits an essentially unique extension to a three-functor formalism D , in which

$$f_! = p_* i_{\sharp} \quad \text{for any factorization } f = p \circ i.$$

Recently, Cnossen–Lenz–Linskens [CLL25] gave a conceptually streamlined account of this extension theorem. They enhance $\text{Span}(C, E)$ to a 2-category whose 2-morphisms encode the adjunction data of $i_{\sharp}$ and p_* in a fully coherent way, and they prove that the inclusion $C^{\text{op}} \hookrightarrow \text{Span}_2(C, E)_{I, P}$ is a universal (I, P) -biadjointable functor. In this setup, Mann’s extension theorem becomes a formal corollary; as a byproduct, the unit and counit witnessing $i_{\sharp} \dashv i^*$ are part of the structure of D , rather than non-canonical choices recoverable only after the fact.

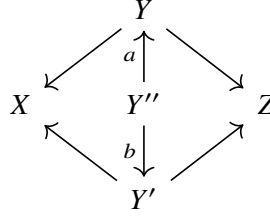
Construction A.1 (CLL 2-categorical span category). The 2-category $\text{Span}_2(C, E)_{I, P}$ has:

- *objects*: objects in C ;
- *1-morphisms* $X \rightarrow Z$: spans

$$\begin{array}{ccc} & Y & \\ f \swarrow & & \searrow g \\ X & & Z \end{array}$$

with $g \in E$;

- 2-morphisms between two such spans: diagrams



with $a \in P$ and $b \in I$.

Theorem A.2 ([CLL25, Theorem 5.17]). *Let C be a category with finite products and let $I, P \subseteq E \subseteq C$ be wide subcategories closed under base change such that*

- (1) *I and P are left cancellable;*
- (2) *every map in E factors as a map in I followed by a map in P ;*
- (3) *every map in $I \cap P$ is truncated.*

Let $D_0: C^{\text{op}} \rightarrow \text{CAlg}(\text{Cat})$ be a functor satisfying:

- (1) *for every $i: a \rightarrow b$ in I , the functor $i^*: D_0(b) \rightarrow D_0(a)$ admits a left adjoint $i_{\sharp}$ satisfying the Beck–Chevalley condition with respect to pullbacks in C and the projection formula $i_{\sharp}(- \otimes i^*(-)) \simeq i_{\sharp}(-) \otimes (-)$;*
- (2) *for every $p: a \rightarrow b$ in P , the functor $p^*: D_0(b) \rightarrow D_0(a)$ admits a right adjoint p_* satisfying the dual Beck–Chevalley condition and dual projection formula;*
- (3) *for every pullback square in C with vertical maps in P and horizontal maps in I , the mixed Beck–Chevalley map $j_{\sharp}p_* \rightarrow q_*i_{\sharp}$ is an equivalence.*

Then the lax symmetric monoidal functor $(C^{\times})^{\text{op}} \rightarrow \text{Cat}^{\times}$ corresponding to D_0 admits a unique extension to a lax symmetric monoidal 2-functor

$$D: \text{Span}_2(C, E)_{I, P}^{\otimes} \longrightarrow \text{Cat}^{\times}.$$

Moreover, for any $e \in E$ with factorization $e = p \circ i$ ($i \in I$, $p \in P$), the exceptional pushforward is $e_! \simeq p_* i_{\sharp}$.

The above is the Mann-style extension formulation; equivalently, the inclusion $C^{\text{op}} \hookrightarrow \text{Span}_2(C, E)_{I, P}$ is the universal (I, P) -biadjointable functor (left base change for I , right base change for P).

Thus, to construct a six-functor formalism it suffices to verify the three biadjointability conditions: left base change for I , right base change for P , and the mixed Beck–Chevalley equivalence. All coherences come for free from the universal property.

Throughout this subsection, (C, E) denotes a geometric setup and $D: \text{Span}(C, E) \rightarrow \text{Cat}$ a three-functor formalism (or a six-functor formalism, when stated). We write Cat_2 for the 2-category of categories.

Cohomological smoothness

The motivating example is Poincaré duality.

Theorem A.3 (Poincaré duality, classical). *Let $f: X \rightarrow Y$ be a smooth submersion of topological manifolds of relative dimension d (i.e. locally of the form $Y \times \mathbb{R}^d \rightarrow Y$). Then $f_!$ admits a right adjoint $f^!$, and there is a sheaf $\omega_f := f^!(\mathbb{1}_Y) \in D(X, \mathbb{Z})$, locally isomorphic to $\mathbb{Z}[d]$, with*

$$f^! \simeq f^* \otimes \omega_f.$$

*The sheaf ω_f is the **dualizing complex** of f . (For f proper one has instead $f_! \simeq f_*$; the two phenomena are independent.)*

A morphism in E which satisfies the analogue of Poincaré duality in D will be called **cohomologically smooth** (or **D-smooth**). Two requirements are immediate:

- There is an isomorphism $f^! \simeq f^* \otimes \omega_f$ for some sheaf ω_f ;
- $\omega_f := f^!(\mathbb{1}_Y)$ is invertible.

But this is not yet enough. Consider a pullback square

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ g' \downarrow & \lrcorner & \downarrow g \\ X & \xrightarrow{f} & Y. \end{array}$$

We expect f' to be D-smooth as well, and the dualizing complexes to be compatible with restriction:

$$g'^* \omega_f \simeq \omega_{f'}.$$

This compatibility is not automatic from pure category theory; it must be imposed.

Definition A.4 (Cohomologically smooth morphism). A morphism $f: X \rightarrow Y$ in E is **cohomologically smooth** (or **D-smooth**) if it satisfies:

(1) The natural transformation

$$f^!(\mathbb{1}_Y) \otimes f^*(-) \longrightarrow f^*(-)$$

is an equivalence; we write $\omega_f := f^!(\mathbb{1}_Y)$ and call it the **dualizing complex** of f .

(2) The dualizing complex ω_f is $\otimes$ -invertible in $D(X)$.

(3) For every pullback square as above with $g \in E$, the morphism f' also satisfies (1)–(2), and the canonical map $g'^* f^!(\mathbb{1}_Y) \rightarrow f'^!(\mathbb{1}_{Y'})$ is an equivalence.

Remark A.5. Cohomological smoothness in the sense of Theorem A.4 is weaker than the geometric notion of a smooth morphism. For instance, in the étale six-functor formalism $X \mapsto \mathrm{Shv}(X_{\text{ét}}, D_{\text{pf}}(\mathbb{Z}))$, every universal homeomorphism between schemes (a map of underlying topological spaces that remains a homeomorphism after pullback) is cohomologically smooth: the étale topology is insensitive to such maps, so they look like isomorphisms at the level of D , and isomorphisms are trivially D -smooth.

The kernel 2-category and Fourier–Mukai transforms

Verifying Theorem A.4 directly is awkward because of the base-change clause (3). The next definitions package D -smoothness into a single 2-categorical statement, after which the base-change clause is automatic.

Fix $Y \in C$, and let $C_E \subseteq C$ be the wide subcategory whose morphisms are those in E . The slice $(C_E)_{/Y}$, paired with the class of *all* morphisms, is itself a geometric setup, and D restricts to a functor

$$D_Y: \mathrm{Span}((C_E)_{/Y}, \mathrm{all})^{\otimes} \rightarrow \mathrm{Span}(C, E)^{\otimes} \xrightarrow{D} \mathrm{Cat}^{\times}.$$

Definition A.6 (Fourier–Mukai transform). Given $X_1, X_2 \in (C_E)_{/Y}$ and $K \in D(X_1 \times_Y X_2)$, the **Fourier–Mukai transform with kernel K** is the functor

$$F_K: D(X_1) \rightarrow D(X_2), \quad A \longmapsto (\pi_2)_!(\pi_1^* A \otimes K).$$

A span $X_1 \xleftarrow{h} Z \xrightarrow{k} X_2$ over Y induces a Fourier–Mukai transform whose kernel is $K = (h, k)_! \mathbb{1}_Z$.

Remark A.7. The terminology is borrowed from the classical analytic transform $T: C^\infty(X) \rightarrow C^\infty(Y)$, $f \mapsto \int_X K(x, y) f(x) dx$, determined by its kernel K . Theorem A.6 categorifies this construction: $\pi_1^* A$ pulls A back to a “function of two variables”, $\otimes K$ multiplies pointwise by the kernel, and $(\pi_2)_!$ is integration along the first variable.

Specializing to $f: X \rightarrow Y$ in E :

- Setting $X_1 = X$, $X_2 = Y$, and $K_1 = \mathbb{1}_X \in D(X \times_Y Y) = D(X)$, the Fourier–Mukai transform is $A \mapsto f_! A$.

- Setting $X_1 = Y$, $X_2 = X$, and $K_2 = \omega_f = f^!(\mathbb{1}_Y) \in \mathbf{D}(Y \times_Y X) = \mathbf{D}(X)$, the transform is $B \mapsto f^*B \otimes \omega_f$.

Idea A.8. The desired identity $f^! \simeq f^* \otimes \omega_f$ should arise by exhibiting the kernels K_1 and K_2 as an adjoint pair in a 2-category whose objects are schemes over Y and whose 1-morphisms are kernels. Since 2-functors preserve adjunctions, the resulting adjunction is automatically compatible with arbitrary base change $g: Y' \rightarrow Y$, taking care of clause (3) of Theorem A.4 for free.

The invertibility of ω_f is a separate, relatively independent condition; this is what motivates the notion of suaveness below.

Definition A.9 (2-category of kernels). Let $Y \in \mathcal{C}$. The **2-category of kernels** $\mathbb{K}_{\mathbf{D},Y}$ is the 2-category whose:

- *Objects* are objects of $(C_E)_{/Y}$, i.e., morphisms $X \rightarrow Y$ in E .
- *1-morphism category* from X_1 to X_2 is $\mathrm{Fun}_Y(X_1, X_2) := \mathbf{D}(X_1 \times_Y X_2)$.
- *Composition* of $M \in \mathrm{Fun}_Y(X_1, X_2)$ and $N \in \mathrm{Fun}_Y(X_2, X_3)$ is

$$N \circ M := (\pi_{13})_!(\pi_{12}^* M \otimes \pi_{23}^* N) \in \mathbf{D}(X_1 \times_Y X_3),$$

where the π_{ij} are the projections from $X_1 \times_Y X_2 \times_Y X_3$.

The identity 1-morphism on X is $\Delta_f!(\mathbb{1}_X) \in \mathbf{D}(X \times_Y X)$, where $\Delta_f: X \rightarrow X \times_Y X$ is the relative diagonal.

Conceptually, $\mathbb{K}_{\mathbf{D},Y}$ is obtained from $\mathrm{Span}((C_E)_{/Y}, \mathrm{all})$, which is closed (being a span category over a category with finite limits) and hence self-enriched, by base-changing the self-enrichment along D_Y ; see [Aok26, Definition 2.4].

There is a canonical 2-functor

$$\Psi_{\mathbf{D},Y} = \mathrm{Fun}_Y(Y, -) : \mathbb{K}_{\mathbf{D},Y} \longrightarrow \mathrm{Cat}_2,$$

sending an object X to $\mathbf{D}(X)$ and a 1-morphism $M \in \mathrm{Fun}_Y(X_1, X_2)$ to the Fourier–Mukai transform $(\pi_2)_!(\pi_1^*(-) \otimes M)$.

For the reader’s convenience, we record three facts about adjunctions in 2-categories that we use freely (proofs are standard, see [HM24, Appendix D]):

- Any 2-functor $F: \mathbb{C} \rightarrow \mathbb{D}$ preserves adjunctions; in fact, it preserves the entire mate correspondence [HM24, Lemma D.2.7].

- Adjunctions are unique up to a contractible space of choices: any two right adjoints to a given 1-morphism are canonically equivalent.
- A pair (f, g) with candidate unit η and counit ε such that the triangle composites are isomorphisms (not necessarily identities) can be corrected to a strict adjunction by modifying η .

Theorem A.10 (Smoothness via kernels). *A morphism $f: X \rightarrow Y$ in E is D-smooth if and only if*

- (1) *The 1-morphism $\mathbb{1}_X \in \text{Fun}_Y(X, Y) = \text{D}(X)$ is a left adjoint in $\mathbb{K}_{\text{D}, Y}$. Denote its right adjoint by $\omega_f \in \text{Fun}_Y(Y, X) = \text{D}(X)$.*
- (2) *ω_f is $\otimes$ -invertible in $\text{D}(X)$.*

When these hold, $\omega_f \simeq f^!(\mathbb{1}_Y)$.

Proof sketch. ($\Rightarrow$) Suppose f is D-smooth. By Theorem A.4(1), $f^!(-) \simeq f^*(-) \otimes \omega_f$. Combined with the adjunction $f_! \dashv f^!$ this becomes $f_! \dashv f^*(-) \otimes \omega_f$ at the level of functors. Translating to $\mathbb{K}_{\text{D}, Y}$, the kernel $\mathbb{1}_X$ corresponds to $f_!$ and ω_f to $f^*(-) \otimes \omega_f$, so $\mathbb{1}_X \dashv \omega_f$ in $\mathbb{K}_{\text{D}, Y}$. Invertibility is clause (2).

($\Leftarrow$) Apply $\Psi_{\text{D}, Y}$ to the adjunction $\mathbb{1}_X \dashv \omega_f$. Since 2-functors preserve adjunctions, the resulting functors $\Psi_{\text{D}, Y}(\mathbb{1}_X) \simeq f_!(-)$ and $\Psi_{\text{D}, Y}(\omega_f) \simeq f^*(-) \otimes \omega_f$ form an adjunction in Cat_2 . By uniqueness of right adjoints, $f^*(-) \otimes \omega_f \simeq f^!(-)$, which is clause (1) of Theorem A.4; setting $- = \mathbb{1}_Y$ gives $\omega_f \simeq f^!(\mathbb{1}_Y)$. The base-change clause (3) follows from the fact that $S \mapsto \mathbb{K}_{\text{D}, S}$ assembles into a 2-functor $\text{Span}(C, E)^\otimes \rightarrow \text{Cat}_2^\times$ [HM24, Theorem 4.2.4], so the adjunction $\mathbb{1}_X \dashv \omega_f$ is preserved by base change. $\square$

Suaveness and primness

Theorem A.10 decomposes cohomological smoothness as “left-adjointness in $\mathbb{K}_{\text{D}, Y}$ ” plus “invertibility of the dualizing complex”. Dropping the invertibility yields broader classes of objects.

Definition A.11 (Suave and prim objects). Let $f: X \rightarrow Y$ be in E and $A \in \text{D}(X)$.

- (1) A is **f -suave** if, viewed as a 1-morphism $X \rightarrow Y$ in $\mathbb{K}_{\text{D}, Y}$, it is a left adjoint. Its right adjoint is the **f -suave dual** $\text{SD}_f(A) \in \text{D}(X)$. Write $\text{Suave}_f(X) \subseteq \text{D}(X)$ for the full subcategory of f -suave objects.
- (2) A is **f -prim** if, viewed as a 1-morphism $Y \rightarrow X$ in $\mathbb{K}_{\text{D}, Y}$, it is a left adjoint. Its right adjoint is the **f -prim dual** $\text{PD}_f(A) \in \text{D}(X)$. Write $\text{Prim}_f(X) \subseteq \text{D}(X)$ for the full subcategory of f -prim objects.

(Equivalently, via the self-duality $\mathbb{K}_{D,Y} \simeq \mathbb{K}_{D,Y}^{\text{op}}$, A is f -prim iff it is a *right* adjoint 1-morphism $X \rightarrow Y$; this is the form used by Heyer–Mann [HM24].)

Remark A.12 (Origin of the terminology). The terms are due to Scholze (suave) and Hansen (prim); applications typically require D to be a six-functor formalism, although the definition makes sense for three-functor formalisms.

- **Suave** (*relaxed smoothness*): when $A = \mathbb{1}_X$ is f -suave, applying $\Psi_{D,Y}$ to the adjunction $\mathbb{1}_X \dashv \text{SD}_f(\mathbb{1}_X)$ gives $f^*(-) \otimes \text{SD}_f(\mathbb{1}_X) \simeq f^!(-)$. More generally, every f -suave A produces a relation between f^* and $f^!$.
- **Prim** (*primitive properness*): when $A = \mathbb{1}_X$ is f -prim, the dual statement gives $f_!(\text{PD}_f(\mathbb{1}_X) \otimes -) \simeq f_*(-)$. Primness is the phenomenon that f_* is recovered from $f_!$ by twisting; this is strictly weaker than the proper-base-change identity $f_! \simeq f_*$, so it is “primitive” in this sense.

In a precise sense, A is f -suave (or f -prim) iff it is “finite-dimensional relative to f ”: it admits a reflexive duality.

Proposition A.13 (Reflexive duality). *Let D be a six-functor formalism on (C, E) and $f: X \rightarrow Y$ in E .*

- (1) *If $A \in D(X)$ is f -suave, then $\text{SD}_f(A)$ is also f -suave (with right adjoint A in $\mathbb{K}_{D,Y}$), and there is a natural isomorphism*

$$\text{SD}_f(A) \otimes f^*(-) \simeq \underline{\text{Hom}}(A, f^!(-)): D(Y) \rightarrow D(X).$$

Setting $- = \mathbb{1}_Y$ yields $\text{SD}_f(A) \simeq \underline{\text{Hom}}(A, f^!(\mathbb{1}_Y))$, exhibiting $\text{SD}_f(A)$ as a relative Verdier dual of A . The bidual identity $\text{SD}_f(\text{SD}_f(A)) \simeq A$ then holds, and SD_f defines a contravariant equivalence

$$\text{SD}_f: \text{Suave}_f(X)^{\text{op}} \xrightarrow{\sim} \text{Suave}_f(X).$$

- (2) *Dually, if $A \in D(X)$ is f -prim, then $\text{PD}_f(A)$ is also f -prim, with*

$$f_!(\text{PD}_f(A) \otimes -) \simeq f_* \underline{\text{Hom}}(A, -): D(X) \rightarrow D(Y),$$

and the bidual $\text{PD}_f(\text{PD}_f(A)) \simeq A$ holds. Consequently PD_f defines a contravariant equivalence

$$\text{PD}_f: \text{Prim}_f(X)^{\text{op}} \xrightarrow{\sim} \text{Prim}_f(X).$$

A more concrete characterization of suaveness and primness is available, useful in computations. It rests on the following standard 2-categorical lemma.

Lemma A.14 (Adjoint in 2-categories, [HM24, Proposition D.2.8]). *Let $\mathbb{C}$ be a 2-category and $f: X \rightarrow Y$ a 1-morphism. The following are equivalent:*

- (1) *f is a left adjoint.*
- (2) *For each $Z \in \{X, Y\}$, the post-composition functor $f_*: \text{Fun}_{\mathbb{C}}(Z, X) \rightarrow \text{Fun}_{\mathbb{C}}(Z, Y)$, $h \mapsto f \circ h$, admits a right adjoint G_Z , and the canonical morphism $G_Y(\text{id}_Y) \circ f \rightarrow G_X(f)$ becomes an isomorphism after applying $\text{Hom}_{\text{Fun}(X, X)}(\text{id}_X, -)$.*
- (3) *For each $Z \in \mathbb{C}$, f_* admits a right adjoint G_Z , and for every $h: Z \rightarrow Z'$ in $\mathbb{C}$, the canonical map $h^* G_{Z'} \rightarrow G_Z h^*$ is an equivalence of functors.*

The right adjoint of f is then $G_Y(\text{id}_Y): Y \rightarrow X$.

Proposition A.15 (Concrete criterion for suave/prim objects). *Let D be a six-functor formalism on (C, E) and $f: X \rightarrow Y$ in E . Write $\pi_i: X \times_Y X \rightarrow X$ for the projections.*

- (1) *$A \in D(X)$ is f -suave iff the natural map*

$$\pi_1^* \underline{\text{Hom}}(A, f^! \mathbb{1}_Y) \otimes \pi_2^* A \longrightarrow \underline{\text{Hom}}(\pi_1^* A, \pi_2^! A)$$

becomes an isomorphism after applying $\text{Hom}_{D(X)}(\Delta_{f!} \mathbb{1}_X, -) \simeq \text{Hom}_{D(X)}(\mathbb{1}_X, \Delta_f^!(-))$. In this case, for every $Z \in (C_E)_Y$ and $M \in D(Z)$,

$$\text{pr}_X^* \text{SD}_f(A) \otimes \text{pr}_Z^* M \xrightarrow{\sim} \underline{\text{Hom}}(\text{pr}_X^* A, \text{pr}_Z^! M) \quad \text{in } D(X \times_Y Z).$$

- (2) *$A \in D(X)$ is f -prim iff the natural map*

$$f_!(\pi_{2*} \underline{\text{Hom}}(\pi_1^* A, \Delta_{f!} \mathbb{1}_X) \otimes A) \longrightarrow f_* \underline{\text{Hom}}(A, A)$$

becomes an isomorphism after applying $\text{Hom}_{D(Y)}(\mathbb{1}_Y, -)$. In this case, $\text{PD}_f(A) \simeq \pi_{2} \underline{\text{Hom}}(\pi_1^* A, \Delta_{f!} \mathbb{1}_X)$, and for every Z, M ,*

$$\text{pr}_{Z!}(\text{pr}_X^* \text{PD}_f(A) \otimes M) \xrightarrow{\sim} \text{pr}_{Z*} \underline{\text{Hom}}(\text{pr}_X^* A, M).$$

Corollary A.16 (Suave–prim duality pairing). *Let $f: X \rightarrow Y$ in E , with $A \in D(X)$ f -suave and $B \in D(X)$ f -prim. Then there is a natural isomorphism*

$$f_* \underline{\text{Hom}}(B, \text{SD}_f(A)) \simeq f_* \underline{\text{Hom}}(\text{PD}_f(B), A)^\vee.$$

Proposition A.17 (Closure properties). *Let $f: X \rightarrow Y$ be in E .*

- *$\text{Suave}_f(X)$ and $\text{Prim}_f(X)$ are closed under retracts in $D(X)$.*
- *For any small $\mathcal{I}$ such that D is compatible with $\mathcal{I}$ -indexed colimits and $\mathcal{I}^{\text{op}}$ -indexed limits, $\text{Suave}_f(X)$ and $\text{Prim}_f(X)$ are closed under such colimits and limits.*

In particular, when D is a stable three-functor formalism, $\text{Suave}_f(X)$ and $\text{Prim}_f(X)$ are closed under retracts, fibers, and cofibers, hence are thick subcategories of $D(X)$.

Suave morphisms and prim morphisms

Specializing Theorem A.11 to $A = \mathbb{1}_X$ yields classes of *morphisms*.

Definition A.18 (Suave and prim morphisms). Let $f: X \rightarrow Y$ be in E .

- (1) f is **D-suave** if $\mathbb{1}_X \in \text{Suave}_f(X)$. We then write $\omega_f := \text{SD}_f(\mathbb{1}_X) \in \text{D}(X)$ for the **dualizing complex** of f .
- (2) f is **D-prim** if $\mathbb{1}_X \in \text{Prim}_f(X)$. We then write $\delta_f := \text{PD}_f(\mathbb{1}_X) \in \text{D}(X)$ for the **co-dualizing complex** of f .

Remark A.19. Combining Theorem A.10 with Theorem A.18 gives

$$f \text{ is D-smooth} \iff f \text{ is D-suave and } \omega_f \text{ is invertible.}$$

Theorem A.13 and Theorem A.15 provide explicit formulas:

$$\omega_f \simeq f^!(\mathbb{1}_Y) \quad \text{and} \quad \delta_f \simeq \pi_{2*} \Delta_{f!} \mathbb{1}_X,$$

where $\pi_2: X \times_Y X \rightarrow X$ is the projection onto the second factor.

For D-suave f , the formula $\omega_f \otimes f^*(-) \xrightarrow{\sim} f^!(-)$ is equivalent (by adjunction) to $f^*(-) \xrightarrow{\sim} \underline{\text{Hom}}(\omega_f, f^!(-))$. Plugging in $\mathbb{1}_Y$ yields $\mathbb{1}_X \xrightarrow{\sim} \underline{\text{Hom}}(\omega_f, \omega_f)$, and for dualizable ω_f this gives $\mathbb{1}_X \simeq \omega_f^\vee \otimes \omega_f$. We record this as an observation.

Observation A.20. For a D-suave morphism f , the dualizing complex ω_f is $\otimes$ -invertible if and only if it is dualizable. Consequently the verification of D-smoothness reduces to checking suaveness and dualizability of ω_f .

We close with the base-change identities for suave and prim morphisms; they will guide the definitions of cohomologically étale and proper morphisms in the next subsection.

Lemma A.21 (Base change for suave and prim morphisms, [HM24, Lemma 4.5.13]). *Let $\mathcal{D}$ be a six-functor formalism on (C, E) and consider a pullback square*

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ g' \downarrow & \lrcorner & \downarrow g \\ X & \xrightarrow{f} & Y \end{array}$$

in C_E .

(1) If f is D-suave, the natural transformations

$$g'_* f'^* \simeq f^* g_*, \quad g'_! f'^! \simeq f^! g_!, \quad f'^* g'^! \simeq g'^! f'^*, \quad g'^* f'^! \simeq f'^! g'^*$$

are all isomorphisms.

(2) If f is D-prim, the natural transformations

$$g^* f_* \simeq f'_* g'^*, \quad g^! f_! \simeq f'_! g'^!, \quad f_! g'_* \simeq g_* f'_!, \quad f_* g'_! \simeq g'_! f_*$$

are all isomorphisms.

Cohomological étaleness and properness

We now ask: under what conditions are there genuine isomorphisms $f^! \simeq f^*$ and $f_! \simeq f_*$, rather than mere twists by ω_f or δ_f ?

A candidate map $f^! \rightarrow f^*$ is constructed using the diagonal. Consider

$$\begin{array}{ccccc} X & & \xrightarrow{\text{id}_X} & & X \\ & \searrow \Delta_f & & \searrow \pi_2 & \\ & X \times_Y X & \xrightarrow{\pi_2} & X & \\ & \searrow \text{id}_X & \pi_1 \downarrow \lrcorner & \downarrow f & \\ & X & \xrightarrow{f} & Y. & \end{array}$$

We have $f\pi_1 = f\pi_2$ and $\text{id}_X = \pi_2\Delta_f$. Suppose inductively that Δ_f satisfies $\Delta_f^! \simeq \Delta_f^*$. Then

$$\begin{aligned} f^! &\simeq \text{id}_X^* f^! \simeq \Delta_f^* \pi_2^* f^! \xrightarrow{\sim} \Delta_f^! \pi_2^* f^! \\ &\rightarrow \Delta_f^! \pi_1^! f^* \simeq (\pi_1 \Delta_f)^! f^* \simeq f^*, \end{aligned}$$

where $\pi_2^* f^! \rightarrow \pi_1^! f^*$ is induced (via right adjoints) from the base-change isomorphism $\pi_{1!} \pi_2^* \simeq f^* f_!$. By Theorem A.21, when f is D-suave the latter is an isomorphism, and so $f^! \simeq f^*$.

A dual construction gives $f_! \rightarrow f_*$, an isomorphism when Δ_f satisfies $\Delta_f^! \simeq \Delta_{f*}$ and f is D-prim. To make the recursion well-founded we impose a truncatedness hypothesis: the recursion terminates at -2 -truncated morphisms (i.e., isomorphisms), where the conclusion is trivial.

Definition A.22 (Cohomologically étale and proper morphisms). Let D be a six-functor formalism on (C, E) , and let $f: X \rightarrow Y$ be a truncated morphism in E (i.e., n -truncated for some $n \in \mathbb{N}$).

- f is **D-étale** (or **cohomologically étale**) if f is D-suave and Δ_f is either D-étale or an isomorphism.

- f is **D-proper** (or **cohomologically proper**) if f is D-prim and Δ_f is either D-proper or an isomorphism.

These definitions are well-founded because n -truncated morphisms have $(n-1)$ -truncated diagonals.

Proposition A.23 (Characterization of étaleness and properness). *Let D be a six-functor formalism on (C, E) and $f: X \rightarrow Y$ in E .*

(1) *Suppose Δ_f is D-étale. There is a natural map $f^! \rightarrow f^*$, and the following are equivalent:*

- (a) f is D-étale;
- (b) $f^! \mathbb{1}_Y \simeq f^* \mathbb{1}_Y$ in $D(X)$;
- (c) $f^! \xrightarrow{\sim} f^*$ as functors.

(2) *Suppose Δ_f is D-proper. There is a natural map $f_! \rightarrow f_*$, and the following are equivalent:*

- (a') f is D-proper;
- (b') $f_! \mathbb{1}_X \simeq f_* \mathbb{1}_X$ in $D(Y)$;
- (c') $f_! \xrightarrow{\sim} f_*$ as functors.

This completes the foundational vocabulary — cohomological smoothness, suaveness, primness, étaleness, properness — whose translation under the transmutation $[-]_D$ is the subject of the dictionary in the main text.

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